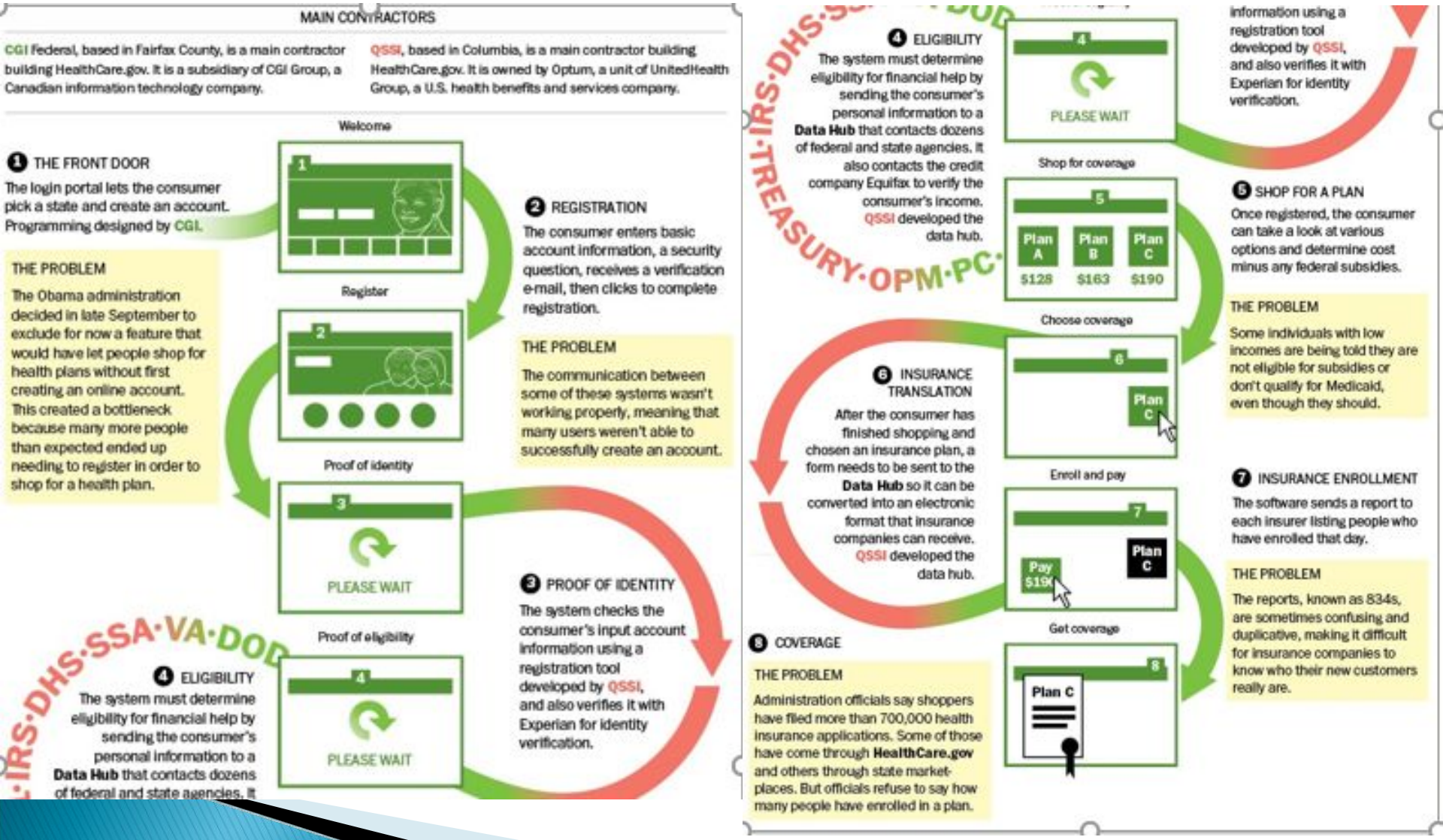
The Healthcare.gov Case



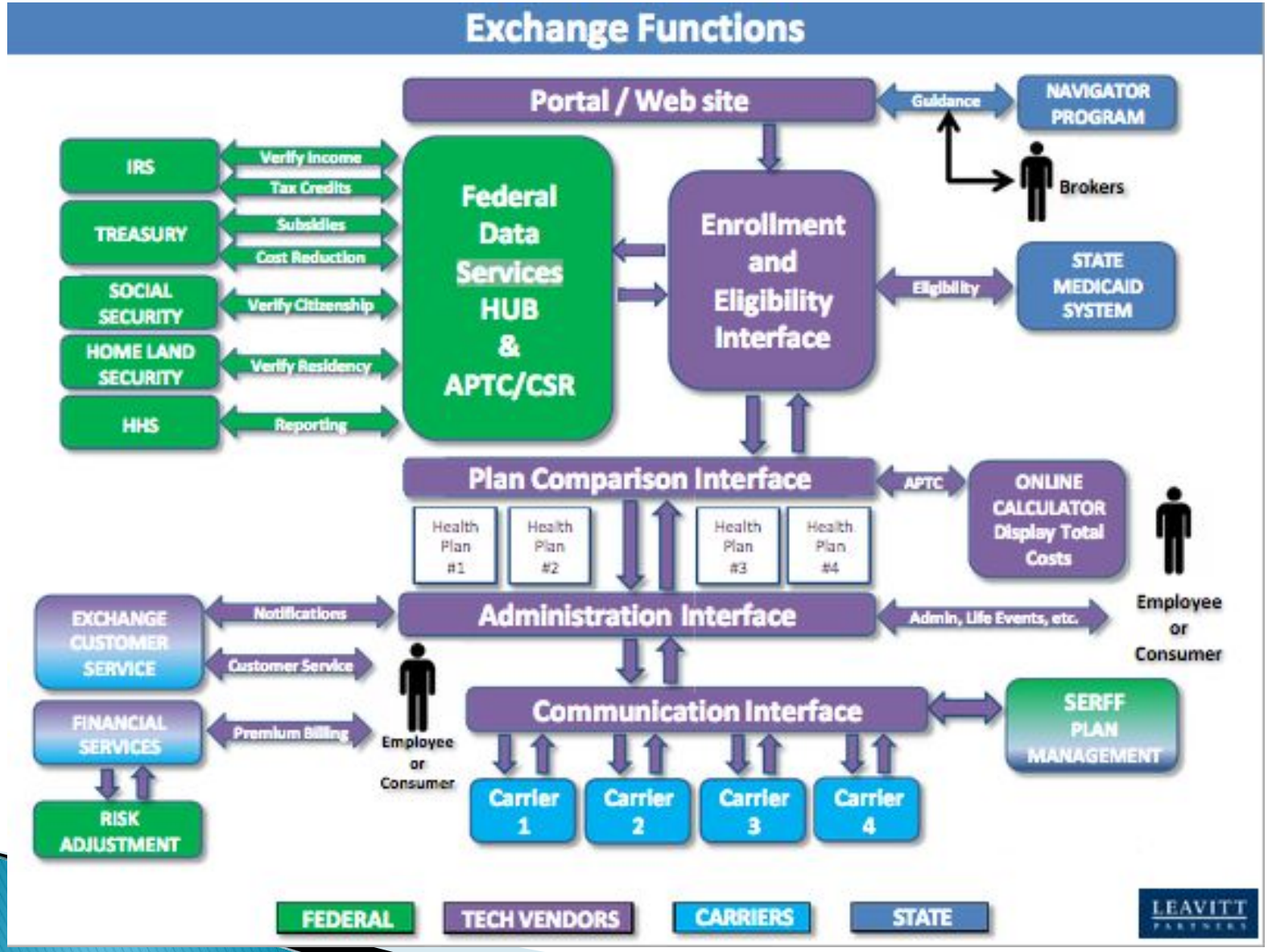
Background

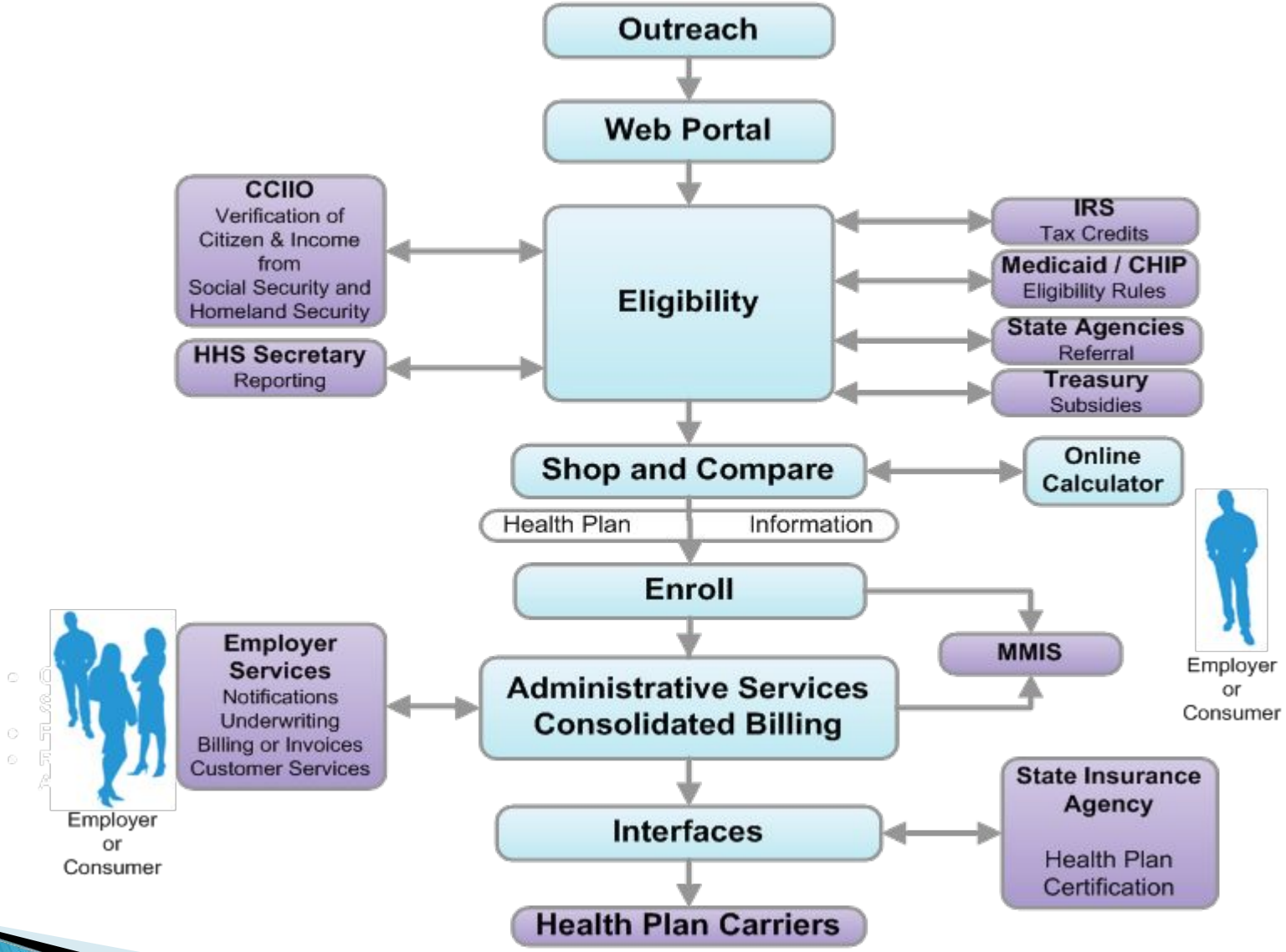
- March 2010: Obama signs Affordable Care Act
- HealthCare.gov is a healthcare exchange website.
  - “One-stop shopping sites for health insurance”
  - CBO forecast: 7 million users during the first year
- Development contracts awarded in September 2011
- HealthCare.gov launched on October 1, 2013
  - Serious technological problems

Process & Problems



Architecture




Outreach
Web Portal
CCIIO
Verification of Citizen & Income from Social Security and Homeland Security
HHS Secretary
Reporting
Eligibility
IRS
Tax Credits
Medicaid / CHIP
Eligibility Rules
State Agencies
Referral
Treasury
Subsidies
Shop and Compare
Online Calculator
Health Plan
Information
Enroll
MMIS
Employer Services
Notifications
Underwriting
Billing or Invoices
Customer Services
Administrative Services
Consolidated Billing
Employer or Consumer
Employer or Consumer
State Insurance Agency
Health Plan Certification
Interfaces
Health Plan Carriers

Launch Problems

- Performance: response time (landing page) > 8s
- Navigation: broken UI
- Stability: intermittent crashes, availability ≈ 43%
- Functionality: incorrect and incomplete data
- Error rate (per page) ≈ 6%
- Scalability: < 1,100 concurrent users
- Enrollment completion rate < 30%

Assignment

- After the launch failure, you have been assigned to a team of architects that has to evaluate the architecture
- You will use the ATAM model to evaluate the architecture
- This evaluation will be used as input for a new version of the system
- Starting point: https://oig.hhs.gov/oei/reports/oei-06-14-00350.asp

Assignment

- Divide in two groups. Each group will simulate ATAM Phase 0 and Phase 1
- What should be done in Phase 0? Review and simulate
- What should be done in Phase 1? Review and simulate
  - Evaluate situation
  - Create scenarios
  - Create utility tree
  - Presente including utility
- Present to the class